

Program 8: create kubernetes pods using YAML descriptors

program-8/

```
|— server.js
|— nodejs-configmap.yaml
|— nodejs-pod.yaml
```

Step 1: Create [server.js](#)

nano server.js

```
const http = require("http");
const port = 3000;

const server = http.createServer((req, res) => {
  res.end("Hello from Node.js running inside Kubernetes!");
});

server.listen(port, () => {
  console.log(`Server running at port ${port}`);
});
```

Step 2: Create ConfigMap YAML

nano nodejs-configmap.yaml

```
apiVersion: v1
kind: ConfigMap
metadata:
  name: nodejs-app-config
data:
  server.js: |
    const http = require("http");
    const port = 3000;

    const server = http.createServer((req, res) => {
      res.end("Hello from Node.js running inside Kubernetes!");
    });

    server.listen(port, () => {
      console.log(`Server running at port ${port}`);
    });
```

Step 3: Create Pod YAML

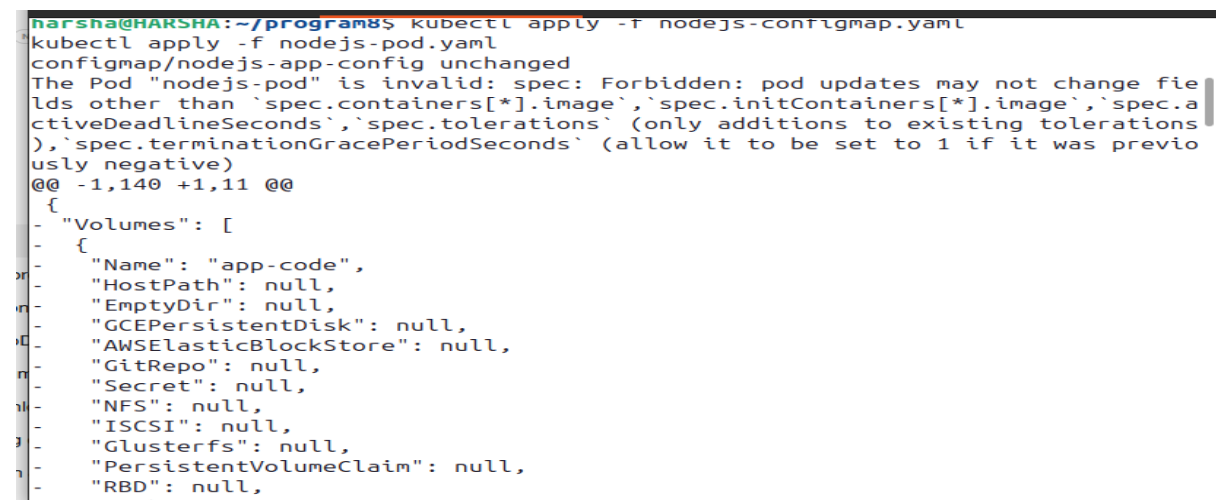
nano nodejs-pod.yaml

```
apiVersion: v1
kind: Pod
metadata:
  name: nodejs-pod
  labels:
    app: nodejs
spec:
  containers:
    - name: nodejs
      image: node:18
      command: ["node", "/usr/src/app/server.js"]
      volumeMounts:
        - name: app-code
          mountPath: /usr/src/app
      ports:
        - containerPort: 3000
  volumes:
    - name: app-code
      configMap:
        name: nodejs-app-config
```

Step 4: Apply the YAML files

kubectl apply -f nodejs-configmap.yaml

kubectl apply -f nodejs-pod.yaml



```
narsha@HARSHA:~/program8$ kubectl apply -f nodejs-configmap.yaml
kubectl apply -f nodejs-pod.yaml
configmap/nodejs-app-config unchanged
The Pod "nodejs-pod" is invalid: spec: Forbidden: pod updates may not change fields other than `spec.containers[*].image`, `spec.initContainers[*].image`, `spec.activeDeadlineSeconds`, `spec.tolerations` (only additions to existing tolerations), `spec.terminationGracePeriodSeconds` (allow it to be set to 1 if it was previously negative)
@@ -1,140 +1,11 @@
@@
{
  "Volumes": [
    {
      "Name": "app-code",
      "HostPath": null,
      "EmptyDir": null,
      "GCEPersistentDisk": null,
      "AWSElasticBlockStore": null,
      "GitRepo": null,
      "Secret": null,
      "NFS": null,
      "ISCSI": null,
      "Glusterfs": null,
      "PersistentVolumeClaim": null,
      "RBD": null,
      ..
    }
  ]
}
```

Step 5: Check Pod Status

kubectl get pods

```
harsha@HARSHA:~/program8$ kubectl get pods
NAME          READY   STATUS    RESTARTS   AGE
nodejs-pod    1/1     Running   4 (99s ago) 6d22h
```

Step 6: Access the Node.js Application

kubectl port-forward pod/nodejs-pod 3000:3000

Open browser:

http://localhost:3000

```
nodejs-pod    1/1     Running   4 (99s ago) 6d22h
harsha@HARSHA:~/program8$ kubectl port-forward pod/nodejs-pod 3000:3000
Forwarding from 127.0.0.1:3000 -> 3000
Forwarding from [::1]:3000 -> 3000
Handling connection for 3000
```

← → ↻   http://localhost:3000 

Hello from Node.js running inside Kubernetes!